DSA210 Term Project Final Report **Dilay Arkın 32288**

Project Title: AI Usage and Academic Performance: A Data-Driven Analysis

Introduction

Artificial intelligence tools such as ChatGPT and Gemini are increasingly used by university students during studying, assignments, and exam preparation. While these tools can improve efficiency and accessibility of information, excessive dependence on AI may also affect productivity and academic performance negatively.

The motivation of this project is to analyze the relationship between AI usage habits and academic productivity using a self-collected dataset. The project aims to investigate whether factors such as AI usage duration, study behavior, course type, and deadlines are associated with changes in productivity and performance scores.

Data Collection & Dataset

The dataset used in this project was self-collected through daily observations between March and May 2026. A total of 60 observations were recorded.

The dataset includes variables related to:
- AI usage duration
- Purpose of AI usage
- Number of AI interactions
- Study duration
- Productivity level
- Task type
- Deadline proximity
- Course type
- Difficulty level
- Performance score

The data was manually recorded in a structured spreadsheet and later analyzed using Python libraries such as Pandas, NumPy, Matplotlib, and Scikit-learn.

EDA Section

Exploratory Data Analysis (EDA) was performed to understand the distribution of variables and relationships within the dataset.

The analysis showed that AI usage varied significantly depending on course type, deadlines, and task difficulty. Higher AI usage was generally observed during assignment-heavy periods and quantitative courses.

Correlation analysis indicated a weak negative relationship between AI usage and productivity. This suggests that increased AI usage may slightly correspond to lower productivity levels, although the relationship is not strong enough to imply causation.

Exploratory Data Analysis was conducted to better understand the dataset structure, variable distributions, and relationships between AI usage and academic productivity.

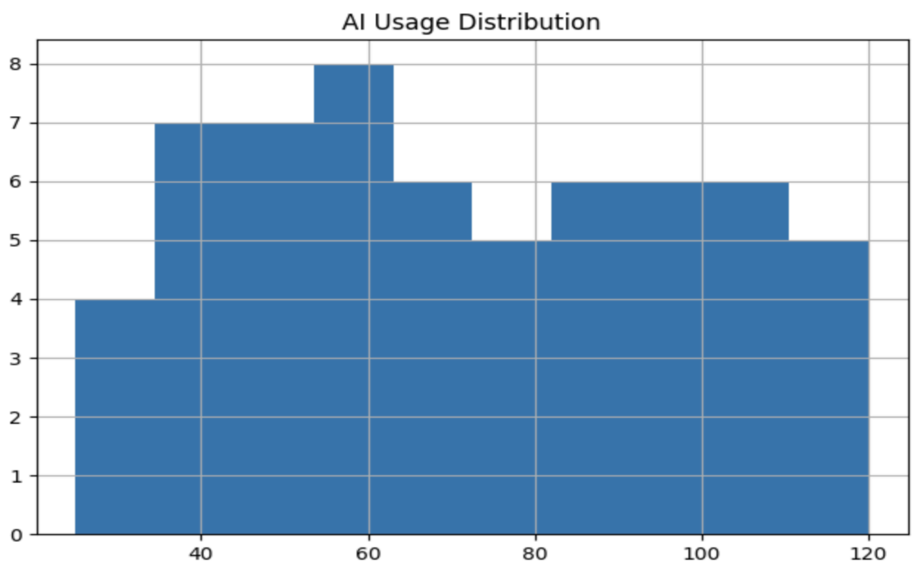

Figure 1 shows the distribution of AI usage duration across the observations.

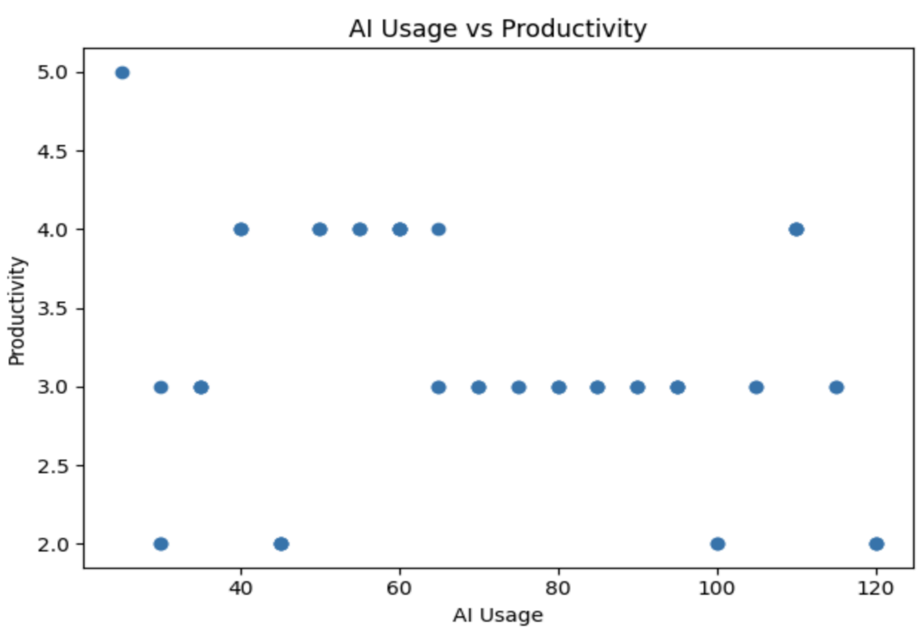

Figure 2 illustrates the relationship between AI usage and productivity levels.

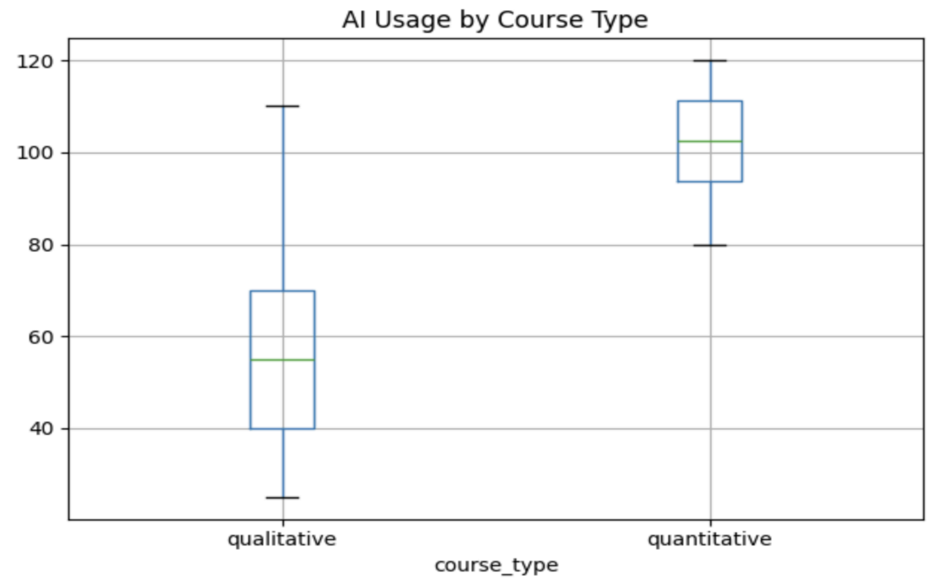

Figure 3 compares AI usage patterns between qualitative and quantitative courses.

The visualizations indicate that AI usage varies depending on course type and academic workload. Quantitative courses generally show higher levels of AI usage. In addition, the relationship between AI usage and productivity appears weakly negative rather than strongly correlated.

Hypothesis Testing

A two-sample t-test was conducted to compare AI usage between quantitative and qualitative courses.

The results showed a statistically significant difference between the two groups ($p < 0.05$), indicating that AI usage patterns differ depending on course type.

However, because the dataset is relatively small and self-collected, the findings should be interpreted cautiously.

Machine Learning Section

- **Regression Analysis**

Regression models were applied to predict performance scores using study-related and AI-related features.

Three supervised learning methods were implemented:
- Linear Regression
- Decision Tree Regressor
- K-Nearest Neighbors (KNN) Regressor

The models achieved relatively high R^2 scores, especially the KNN regression model. However, the dataset size is limited, and the high performance may be partly due to overfitting.

- **Classification**

Classification models were used to predict whether a student has high productivity.

The following methods were implemented:
- Logistic Regression
- Decision Tree Classifier
- KNN Classifier

The models achieved moderate classification accuracy. Logistic Regression produced high recall but relatively low precision, suggesting that the model tends to over-predict high productivity cases.

Limitations

This project has several limitations. First, the dataset was self-collected and relatively small compared to large-scale academic datasets. Although the number of observations was increased during the project, the dataset may still not fully represent real student behavior patterns.

Second, some variables, such as productivity, were partially subjective and based on self-evaluation. In addition, the strong model performances observed in some machine learning tasks may indicate overfitting due to the limited dataset size.

Future Work

Future work could include collecting data from multiple students over a longer period of time to improve generalizability.

Additional variables such as stress levels, sleep duration, GPA, and exam results could also be incorporated to create more robust predictive models.

More advanced machine learning methods and cross-validation techniques could further improve model evaluation and reliability.

Conclusion

Overall, this project successfully applied data science and machine learning techniques to analyze the relationship between AI usage and academic performance.

The findings suggest that AI usage alone is not a sufficient predictor of productivity or academic success. Instead, academic outcomes appear to be influenced by multiple interacting factors such as study duration, deadlines, task type, and course difficulty.

The project demonstrates how self-collected behavioral data can be analyzed using statistical and machine learning methods within a complete data science workflow.

AI Tool Disclosure

AI tools, including ChatGPT, were used during the development of this project for guidance in organization, support, refinement, and assistance. All final analyses, dataset preparation, and implementation decisions were reviewed and validated by the me.